Reliability Assessment of the JAXA H3 Fairing under Manufacturing Variability Using Polynomial Chaos Expansion as a Stochastic FEM Surrogate

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Abstract

This study applies non-intrusive Polynomial Chaos Expansion (PCE) with Smolyak sparse quadrature as a Stochastic Finite Element Method surrogate to quantify how manufacturing variability in five material parameters propagates to the structural response of the JAXA H3 rocket fairing, a CFRP/aluminium-honeycomb sandwich structure modelled in Abaqus/Standard. A degree-2 expansion built from 66 simulations reproduces the response statistics of a degree-3 expansion costing 286 simulations to four significant figures, identifying degree 2 as the correct model order rather than an economy measure. Sobol analysis shows that the fibre-direction modulus E_1 accounts for 99.2% of the variance in peak von Mises stress and 99.95% of the variance in maximum displacement, while the three cohesive-zone parameters are structurally unidentifiable because the adhesive never damages at the design load. The peak stress is 209.7 ± 2.38 MPa against a 1,200 MPa strength, a margin of 416 standard deviations. Three extensions build on the same surrogate: Bayesian identification of E_1 from stress measurements, a non-Gaussian translation random field constructed on a Karhunen–Loève basis, and a systematic comparison of stochastic and data-driven surrogates as a function of training-set size.

Keywords: Polynomial chaos expansion, Stochastic finite element method, Sobol sensitivity indices, Reliability analysis, Karhunen–Loève expansion, CFRP composite structures

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1. Introduction

Composite fairing structures for launch vehicles are manufactured from CFRP skins bonded to aluminium honeycomb cores. The properties of both the laminate and the adhesive interface vary from panel to panel because of fibre volume fraction fluctuations, cure cycle variations and adhesive film thickness inconsistencies. These uncertainties propagate into stress, damage and deformation, and therefore into structural reliability. A deterministic analysis at nominal properties cannot say how much of the computed margin is genuine and how much is an artefact of the particular property values assumed.

Stochastic Finite Element Methods provide the framework for propagating such input uncertainty through a deterministic model. Ghanem and Spanos [1] introduced the spectral approach, expanding the response in Hermite polynomials of Gaussian variables and converting the stochastic boundary-value problem into a deterministic system. That formulation is intrusive: it requires rewriting the finite-element equations, which is not possible with a commercial solver. Sudret and Der Kiureghian [2] review the non-intrusive alternative, in which the deterministic code is treated as a black box and the expansion coefficients are recovered by regression or quadrature from a set of ordinary solver runs. Sudret [3] showed that once such a surrogate exists, Sobol sensitivity indices follow analytically from its coefficients at no additional cost — the property that makes PCE attractive when sensitivity, and not only response statistics, is the object of the study. The polynomial family is dictated by the input measure through the Wiener–Askey scheme [4]; Gaussian inputs select the Hermite basis used here.

The cost of a non-intrusive expansion is governed by the quadrature rule. Full tensor-product Gauss–Hermite quadrature requires n_q^d points, which is prohibitive beyond a few dimensions. The Smolyak sparse grid [5] combines low-order one-dimensional rules so that exactness is retained on the total-degree polynomial space at a fraction of the point count, and sparse expansions can be pushed further with adaptive designs [6]. Spatial variability, when it matters, is represented by the Karhunen–Loève expansion [7, 8], and non-Gaussian marginals are recovered from a Gaussian basis by memoryless translation [9]. On the inverse side, Tarantola [10] sets out the Bayesian framework and Marzouk et al. [11] demonstrate that a PCE surrogate serves as an efficient likelihood evaluator. For the data-driven comparison, Rasmussen and Williams [12] and Forrester et al. [13] provide the reference treatments of Gaussian Processes and of surrogate modelling in engineering design.

This work applies non-intrusive PCE to an existing high-fidelity Abaqus/Standard model of an H3 fairing sandwich panel. The objectives are to construct and validate PCE surrogates of degree 2 and 3 on Smolyak sparse grids; to identify the dominant uncertain parameter by Sobol analysis; to assess reliability under the design load; and to extend the analysis in three directions — Bayesian identification of the dominant parameter from response measurements, a non-Gaussian spatially varying representation of that parameter, and a systematic comparison of stochastic against data-driven surrogates under a small data budget. Fig. 1 summarises the workflow.

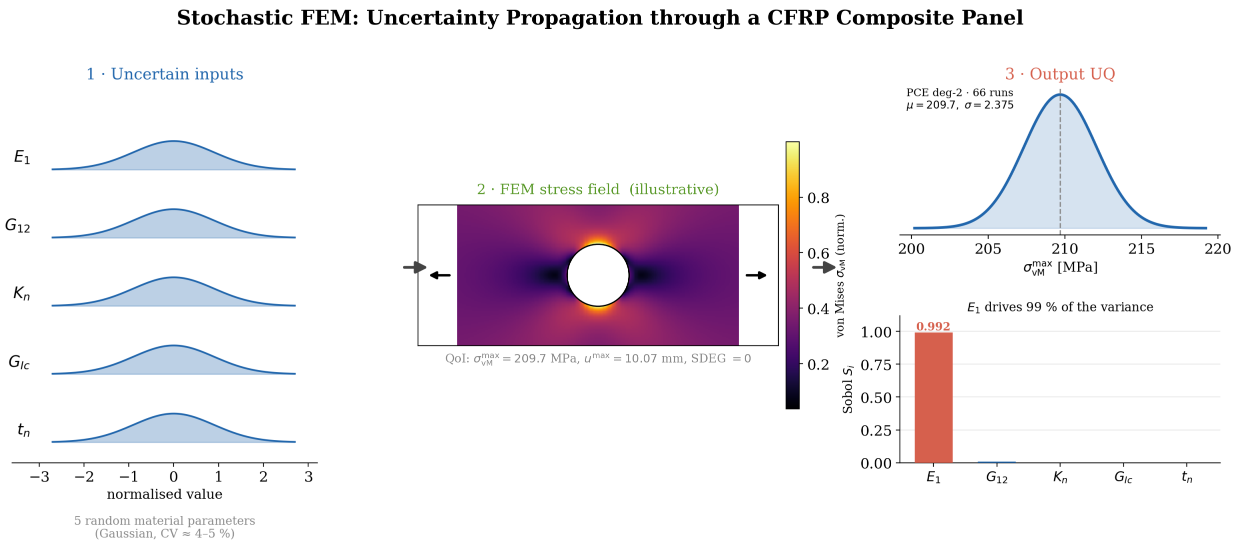


Figure 1: The non-intrusive SFEM workflow: five uncertain material parameters enter as standardised Gaussian inputs, are propagated through the deterministic Abaqus model at the sparse-grid nodes, and yield the response distribution and Sobol indices from the fitted expansion.

2. Methodology

2.1. Polynomial chaos expansion

For a response $Y = \mathcal{M}(\xi)$ depending on a random input vector $\xi \in \mathbb{R}^d$, the expansion reads

$$Y \approx \hat{Y} = \sum_{\alpha \in \mathcal{A}} c_\alpha \Psi_\alpha(\xi), \tag{1}$$

where the Ψ_α are multivariate polynomials orthogonal with respect to the input measure and $\mathcal{A} = \{\alpha : |\alpha| \leq p\}$ is the total-degree truncation. The number of terms is $P = \binom{d+p}{p}$, giving $P = 21$ for $d = 5,\ p = 2$ and $P = 56$ for $p = 3$.

The Hermite basis is not an arbitrary choice but is fixed by the Gaussian input measure, and it is the resulting orthogonality — not the polynomial form itself — that carries the practical value. Because $\mathbb{E}[\Psi_\alpha \Psi_\beta] = 0$ for $\alpha \neq \beta$, the coefficients are the spectral projections of the response, so the mean is read directly from c_0 and the variance from $\sum_{\alpha \neq 0} c_\alpha^2\, \mathbb{E}[\Psi_\alpha^2]$. The Sobol indices [14] follow by the same mechanism as partial sums of squared coefficients grouped by the variables each multi-index activates:

$$S_i = \frac{\sum_{\alpha \in \mathcal{A}_i} c_\alpha^2\, \mathbb{E}[\Psi_\alpha^2]}{\sum_{\alpha \neq 0} c_\alpha^2\, \mathbb{E}[\Psi_\alpha^2]}, \qquad \mathcal{A}_i = \{\alpha : \alpha_i > 0,\ \alpha_{j \neq i} = 0\}. \tag{2}$$

The complete second-moment and global-sensitivity information is thus recovered analytically, without a single additional model evaluation. This is the decisive advantage over a generic regression surrogate.

2.2. *Smolyak sparse quadrature*

Full tensor-product quadrature requires $3^5 = 243$ points at degree 2 and $4^5 = 1024$ at degree 3. The Smolyak construction reduces these to 66 and 286 respectively, a factor of roughly 3.6 in both cases, while retaining exactness on the total-degree space. Negative quadrature weights, which can arise in sparse Gauss–Hermite rules, were handled by falling back to least-squares regression (`chaospy.fit_regression`); with 66 points against 21 unknowns the system is comfortably overdetermined, which also permits leave-one-out cross-validation of the fit.

2.3. *Uncertain inputs and quantities of interest*

Five manufacturing-related parameters are treated as independent truncated normals, $\xi_i \sim \mathrm{TruncNormal}(\mu_i, \sigma_i, 0.5\mu_i, 1.5\mu_i)$, listed in Table 1. Dependent properties are scaled proportionally: $E_2 = 10000\,(E_1/\mu_{E_1})$, $G_{23} = 3000\,(G_{12}/\mu_{G_{12}})$, $K_s = K_n/2$ and $G_{IIc} = G_{Ic}/0.3$.

A truncated normal was preferred over an unbounded Gaussian because stiffnesses and fracture energies are strictly positive, and because material accepted for flight hardware is screened against

Table 1: Uncertain input parameters. Truncation at $[0.5\mu, 1.5\mu]$ corresponds to $\pm 10\sigma$ for E_1 and $\pm 2.5\sigma$ for the widest-scattered parameter G_{Ic}, so the bounds are practically inactive in every case.

Parameter	Description	Nominal μ	CoV	σ
E_1	CFRP fibre-direction Young's modulus	146,000 MPa	5%	7,300 MPa
G_{12}	CFRP in-plane shear modulus	5,200 MPa	10%	520 MPa
K_n	Cohesive normal stiffness	1.0×10^6 N/mm^3	15%	1.5×10^5 N/mm^3
G_{Ic}	Mode-I fracture energy	0.3 N/mm	20%	0.06 N/mm
t_n	Cohesive normal strength	50 MPa	15%	7.5 MPa

incoming-inspection limits, so extreme outliers do not represent the qualified population. The assumed coefficients of variation follow the usual ordering for CFRP: the fibre-dominated modulus is the most tightly controlled, while matrix- and interface-dominated quantities scatter more.

Three responses are extracted from the last frame of each analysis through the `odbAccess` interface: the peak von Mises stress in the CFRP skin, with a failure threshold of 1,200 MPa; the peak scalar damage variable SDEG in the adhesive, with a threshold of 0.5; and the peak nodal displacement.

2.4. *Non-intrusive pipeline*

The pipeline treats Abaqus/Standard as a black box. Smolyak nodes are generated in standard-normal space; for each node the material cards of the input deck are patched and the job is run; the responses are extracted and the coefficients fitted; and the post-processing then yields Sobol indices analytically, failure probabilities by Monte Carlo on the surrogate (10^6 samples), and leave-one-out errors. No solver source code is touched at any stage, which is what makes the approach usable with a commercial code.

3. Finite element model

The model, shown in Fig. 2, represents a panel of the fairing sandwich structure. The CFRP skins use an orthotropic LAMINA elastic definition parameterised by E_1, E_2, ν_{12}, G_{12}, G_{13} and

G_{23}; the aluminium honeycomb core is a homogenised linear-elastic continuum rather than a resolved cellular geometry. The adhesive interface is represented by a cohesive zone model with a traction–separation law that is linear elastic up to the strength t_n and then softens under the control of the fracture energy G_{Ic}, with mixed-mode evolution governed by the Benzeggagh–Kenane criterion [15] at $\eta = 2.284$. Loading is applied in two steps, a static pressure followed by an incremental load. Each output database is approximately 293 MB, which is what makes the sample count, rather than the individual solve, the binding constraint on the study. Fig. 3 shows a deterministic solve of the full fairing for context.

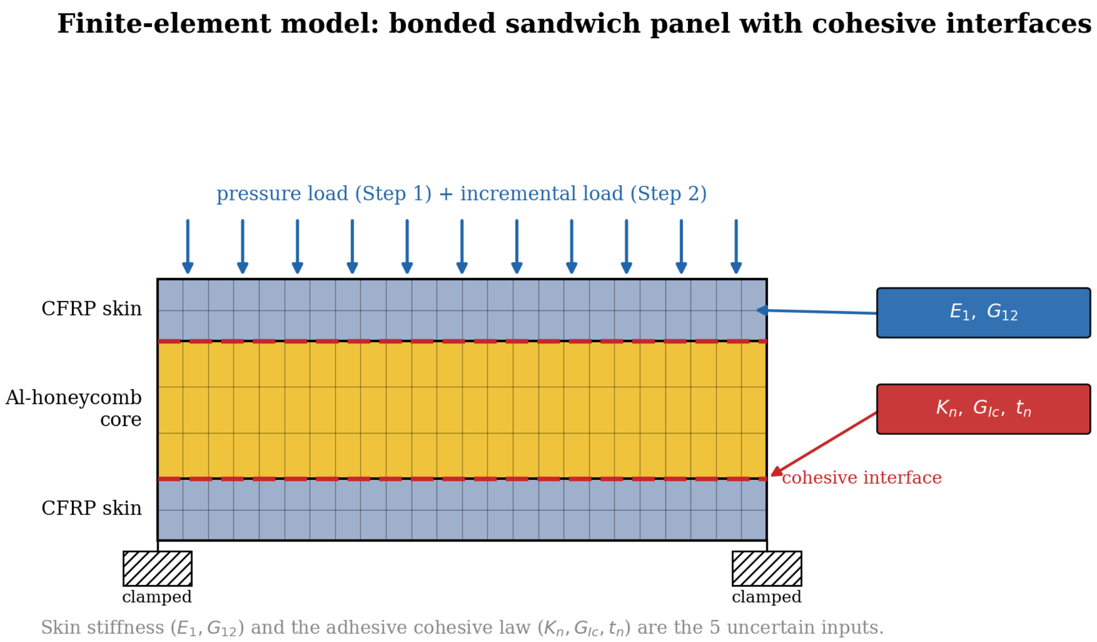


Figure 2: The deterministic model: a CFRP/aluminium-honeycomb sandwich panel with cohesive-zone adhesive interfaces, clamped edges and a two-step pressure and incremental load. The five uncertain inputs enter through the skin (E_1, G_{12}) and the cohesive law (K_n, G_{Ic}, t_n).

4. Forward uncertainty quantification

4.1. Response statistics and convergence

Table 2 collects the response statistics at both expansion degrees. They agree to four significant figures, which establishes convergence at degree 2 and, more informatively, quantifies the

Figure 3: Deterministic solve of the full fairing: von Mises stress concentrating around the access-door cutout (left) and the displacement field peaking at the nose (right). Each quadrature node of the stochastic study is one such black-box solve.

curvature of the response surface: had the responses depended appreciably on higher-order terms, the additional 220 simulations of the degree-3 grid would have moved the standard deviation. That they do not confirms that a quadratic basis already saturates the achievable accuracy; Fig. 4 shows the corresponding convergence with quadrature-point count.

Table 2: PCE response statistics at degree 2 (66 simulations) and degree 3 (286 simulations).

	Degree 2		Degree 3		
QoI	Mean	Std. dev.	Mean	Std. dev.	CoV
$\sigma_{\mathrm{vM}}^{\max}$	209.7 MPa	2.375 MPa	209.7 MPa	2.378 MPa	1.13%
$d^{\max}$ (SDEG)	0.000	0.000	0.000	0.000	—
$u^{\max}$	10.07 mm	0.178 mm	10.07 mm	0.178 mm	1.76%

4.2. Global sensitivity

The Sobol decomposition in Table 3 reveals a strongly anisotropic sensitivity structure rather than a diffuse spread of influence. The fibre-direction modulus alone explains 99.21% of the stress variance and 99.95% of the displacement variance, and no other parameter reaches the

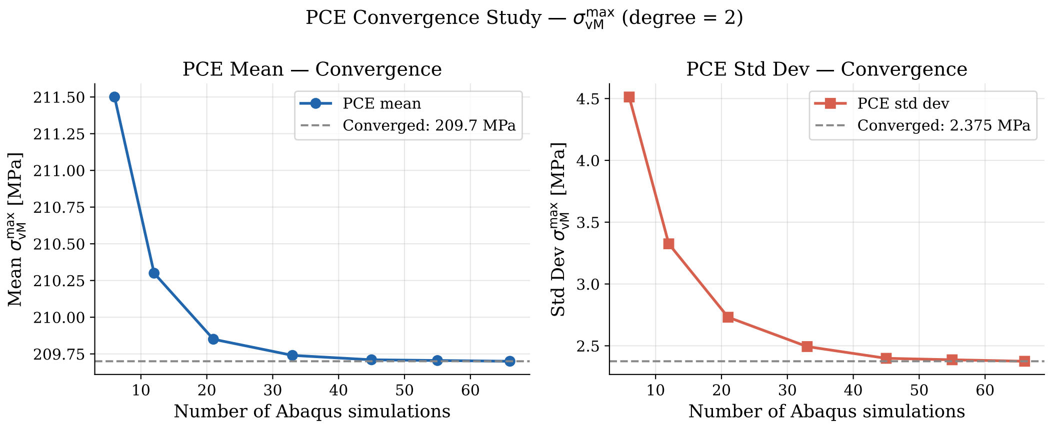


Figure 4: Convergence of the mean and standard deviation of the peak von Mises stress with the number of quadrature points. Both statistics are stable by 66 simulations.

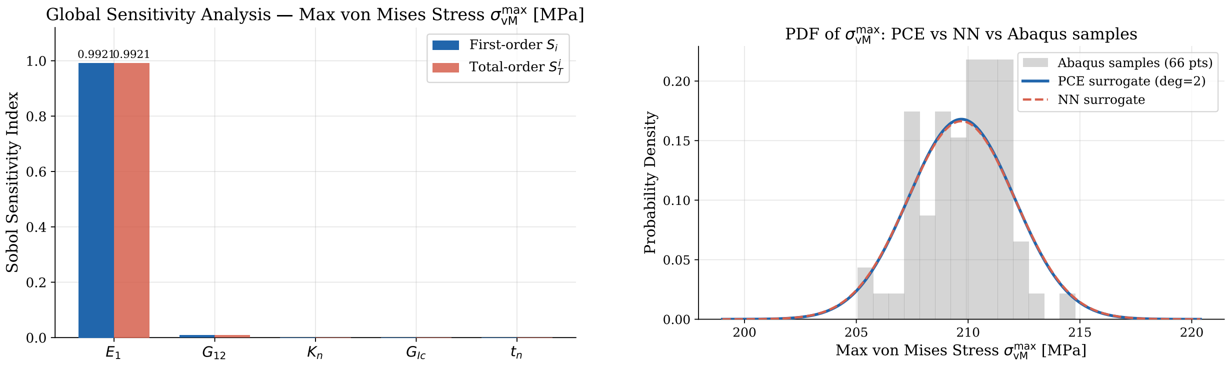
percent level. The shear modulus is the only secondary contributor, and it accounts for under 1% of the stress variance despite carrying twice the coefficient of variation of E_1. The three cohesive parameters register indices between 10^{-11} and 10^{-6}, numerically indistinguishable from zero. The near-exact equality of first- and total-order indices completes the picture: the inputs act additively with no resolvable interactions, the signature of a response that is essentially linear over the sampled range. Fig. 5 shows the attribution alongside the recovered response distribution.

Table 3: First-order and total-order Sobol indices at degree 2.

Parameter	S_i (stress)	S_T^i (stress)	S_i (disp.)	S_T^i (disp.)
E_1	**0.9921**	**0.9921**	**0.9995**	**0.9996**
G_{12}	0.0079	0.0079	0.0005	0.0005
K_n	≈ 0	≈ 0	≈ 0	≈ 0
G_{Ic}	≈ 0	≈ 0	≈ 0	≈ 0
t_n	≈ 0	≈ 0	≈ 0	≈ 0

4.3. Reliability

For a limit state $F = \{G(\xi) > b\}$ the failure probability is $P_f = \int_F f_\xi(x)\,\mathrm{d}x$, estimated by Monte Carlo on the surrogate, and the reliability index is $\beta = -\Phi^{-1}(P_f)$. With a mean stress



(a) Sobol indices for the peak von Mises stress.

(b) Response distribution of the peak stress.

Figure 5: Sensitivity attribution and the recovered response distribution at degree 2.

of 209.7 MPa, a standard deviation of 2.38 MPa and a threshold of 1,200 MPa, sampling 10^6 surrogate realisations returns no failures, so $P_f = 0$ and $\beta = \infty$ at both expansion degrees.

This must be read as "below the resolution of the analysis" rather than as a literal guarantee. Monte Carlo on 10^6 samples resolves probabilities only to $O(10^{-6})$, and the surrogate is valid only within the truncated input domain, outside which the polynomial must not be extrapolated. The defensible statement is that the mean lies $(1200 - 209.7)/2.38 \approx 416$ standard deviations below the threshold, so failure is far outside the modelled range. Fig. 6 places the response distribution against the limit state.

5. Extensions

5.1. Bayesian identification of E_1

That E_1 explains more than 99% of the output variance has a direct inverse implication: among the five parameters it is also the most identifiable from response measurements. Let $\theta = E_1$, with the prior of Table 1 and a linear-Gaussian likelihood obtained from the surrogate,

$$y_j = c_0 + c_1\,\xi_{E_1} + \varepsilon_j, \qquad \varepsilon_j \sim \mathcal{N}(0, \sigma^2_{\text{meas}}), \tag{3}$$

with $c_0 = 209.7$ MPa, $c_1 = 2.365$ MPa and $\sigma_{\text{meas}} = 2.0$ MPa. Because the prior is Gaussian to within a truncation at $\pm 10\sigma$ and the likelihood is linear-Gaussian in ξ_{E_1}, the posterior is Gaussian

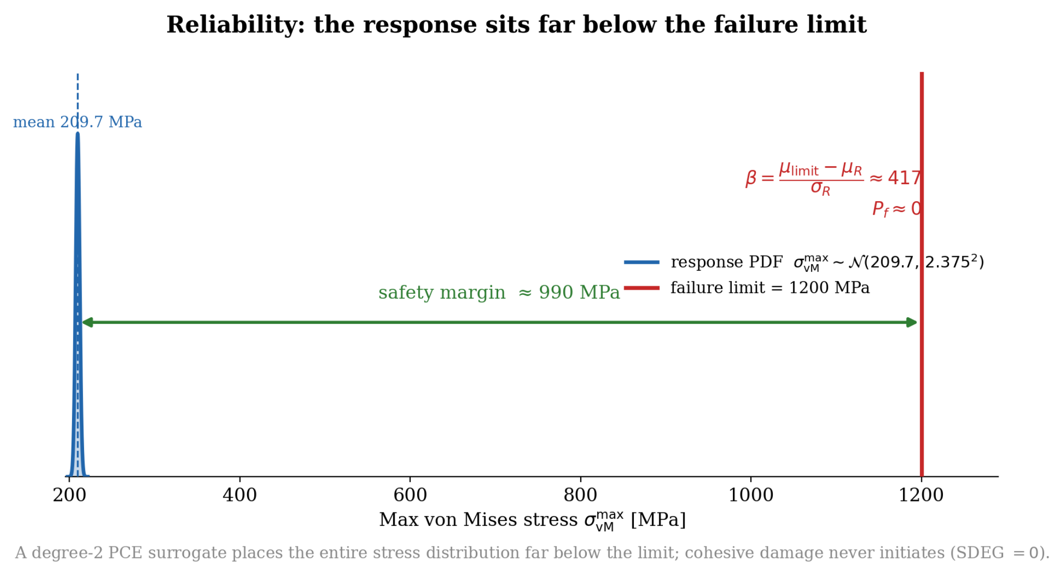


Figure 6: Limit-state view. The entire response distribution lies far below the 1,200 MPa threshold, giving a margin of approximately 990 MPa. Cohesive damage never initiates.

in closed form,

$$\sigma_{\text{post}}^2 = \frac{1}{1 + c_1^2/\sigma_{\text{eff}}^2}, \qquad \mu_{\text{post}} = \sigma_{\text{post}}^2 \frac{c_1(\bar{y} - c_0)}{\sigma_{\text{eff}}^2}, \qquad \sigma_{\text{eff}}^2 = \frac{\sigma_{\text{meas}}^2}{k}, \tag{4}$$

where $\bar{y}$ is the mean of k independent measurements. No sampling is required. In a synthetic experiment with a true E_1 of 153,000 MPa, five measurements reduce the posterior standard deviation from the 7,300 MPa prior to 2,582 MPa, ten measurements to 1,885 MPa — a 74% reduction — and twenty to 1,356 MPa, all at negligible cost once the surrogate exists (Fig. 7).

The two-dimensional likelihood makes the limit of this explicit. Its ridge is almost parallel to the $\xi_{G_{12}}$ axis, with slope $-c_1/c_2 \approx -11$: moving G_{12} across its entire prior range shifts the predicted stress by only 0.21 MPa, well inside the 2.0 MPa measurement noise, so stress observations are essentially uninformative about it. Identifying G_{12} would require roughly 270 measurements, an order-of-magnitude improvement in instrumentation, or — the appropriate remedy — a shear-driven load case in which the parameter actually moves the observable. Identifiability is a property of the experiment, not only of the instrument.

It is worth being precise about what this reduces. The quantity being narrowed is epistemic: the true modulus of a given batch is fixed but unknown, and measurement reduces that ignorance.

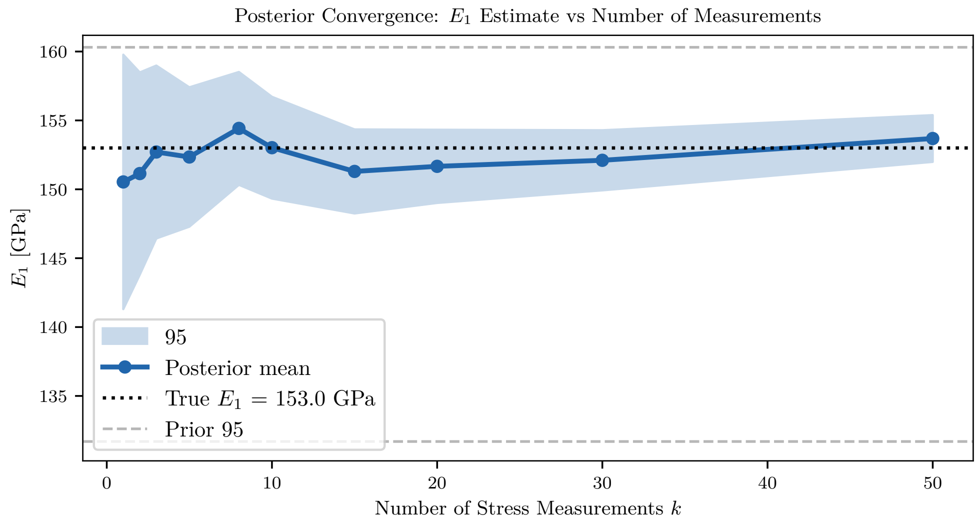


Figure 7: Posterior mean and 95% credible interval of E_1 against the number of stress measurements. Each point uses the mean of k independent observations, so the estimator is consistent and the interval contracts as $\sigma_{\text{meas}}/\sqrt{k}$.

The aleatory scatter between batches is untouched. Since P_f is already zero at a 416-standard-deviation margin, the update does not improve the reliability assessment in any practical sense; its value lies on the manufacturing side, in showing that the parameter which dominates the variance is also the one that cheap response measurements can pin down, which is precisely what tolerance allocation needs.

5.2. *Non-Gaussian random field representation of E_1*

The forward study treats E_1 as a single random variable, which is the $l_c \to \infty$ limit of a random field — appropriate when the dominant variability is between co-cured panels rather than within one. This section relaxes that assumption.

A zero-mean unit-variance Gaussian field $Z(x)$ with covariance $C(x_1, x_2)$ admits the Karhunen–Loève representation

$$Z(x) = \sum_{i=1}^{M} \sqrt{\lambda_i}\, \phi_i(x)\, \zeta_i, \qquad \zeta_i \sim \mathcal{N}(0, 1) \text{ independent}, \tag{5}$$

where (λ_i, ϕ_i) solve the Fredholm integral equation $\int_D C(x_1, x_2)\phi_i(x_2)\, \mathrm{d}x_2 = \lambda_i \phi_i(x_1)$. An anisotropic exponential kernel is adopted,

$$C(x_1, y_1; x_2, y_2) = \exp\left(-\frac{|x_1 - x_2|}{l_{cx}} - \frac{|y_1 - y_2|}{l_{cy}}\right), \tag{6}$$

on a 600×400 mm panel, for short ($l_{cx} = 200$, $l_{cy} = 150$ mm), medium (400, 300) and long (1200, 800) correlation lengths. The eigenpairs are obtained by direct eigendecomposition of the covariance matrix sampled on a uniform 20×15 grid, a point-collocation discretisation of Eq. (5) rather than a Galerkin projection; on a uniform grid the two differ only by the constant cell measure, which cancels in the explained-variance ratios. Reaching 99% of the variance requires 164 modes at the short correlation length, 95 at the medium and 38 at the long: the shorter the correlation, the slower the eigenvalue decay and the higher the effective stochastic dimension.

A Gaussian field assigns non-zero probability to negative stiffness, which is not physically admissible. The standard remedy is a translation field [9], obtained by passing the Gaussian field through a memoryless monotone map,

$$H(x) = F^{-1}(\Phi(Z(x)))\,, \tag{7}$$

which for a lognormal target reduces to $H(x) = \exp(\mu_{\ln} + \sigma_{\ln} Z(x))$ with $\sigma_{\ln}^2 = \ln(1 + \mathrm{CoV}^2)$. Positivity then holds by construction. The transform is monotone, so it preserves rank correlation, but it does *not* preserve linear correlation: the induced correlation is

$$\rho_H = \frac{\exp(\sigma_{\ln}^2 \rho_Z) - 1}{\exp(\sigma_{\ln}^2) - 1}, \tag{8}$$

so if a target ρ_H is prescribed the Gaussian kernel must be pre-distorted to match it.

The magnitude of that distortion is governed by the coefficient of variation, and for the present material it is small. At CoV = 5% the underlying $\sigma_{\ln} = 0.0500$, the marginal skewness is 0.150 and the correlation distortion never exceeds 3.1×10^{-4}; the lognormal and Gaussian descriptions are then numerically almost interchangeable, and the practical gain is the guarantee of positivity rather than any change in the statistics. The distortion grows to 1.1×10^{-2} at CoV = 30% and 3.8×10^{-2} at 60%, so the non-Gaussian treatment becomes material for the interface parameters, whose scatter is larger, rather than for E_1. Both effects are shown in Fig. 8.

Finally, spatial averaging attenuates the variability of the effective panel modulus. The standard deviation of the spatial mean $\overline{E_1}$ falls from the 7,300 MPa point value to approximately 4,700 MPa

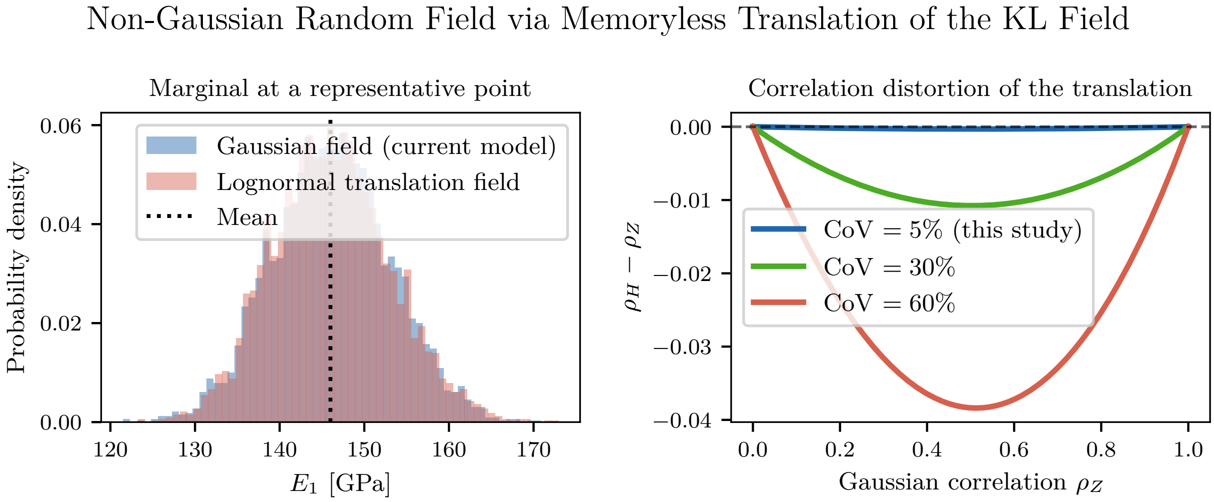


Figure 8: Left: the marginal at a representative point under the Gaussian model and under the lognormal translation field, which is strictly positive by construction. Right: the departure of the induced correlation from the Gaussian one, Eq. (8), against the coefficient of variation.

at the medium correlation length, and further at shorter correlations (Fig. 9). This is an *input-side* result obtained without any additional FE evaluation, and its consequence for the response must be stated carefully. For quantities governed by the panel-average stiffness — global deflection and the membrane response — the homogeneous scalar model over-states the input variability and therefore the response variance, so the scalar treatment is conservative. The same does *not* follow for the peak von Mises stress, which is a local extreme: a heterogeneous field introduces stiffness gradients and stress redistribution that can raise the peak even as the average modulus varies less. Settling that question requires propagating mapped realisations through the FE model, which is beyond the present scope.

5.3. Stochastic against data-driven surrogates

The final extension asks how the choice of surrogate interacts with the size of the training set, which is the operative constraint when each sample is a 293 MB Abaqus solve. PCE of degree 1 and 2, a Gaussian Process with an RBF kernel and a neural network ($5 \to 64 \to 64 \to 32 \to 1$) were trained on $N \in \{10, \ldots, 286\}$ points and evaluated on a fixed test set of 20,000 points, with results reported as medians over five random seeds. At $N = 66$ the test RMSE is 8.9×10^{-3} MPa

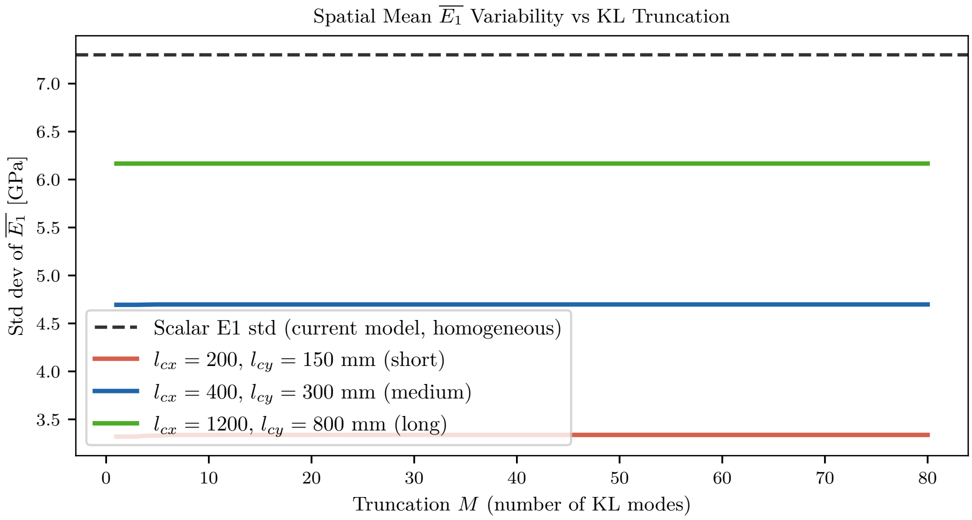


Figure 9: Standard deviation of the spatially averaged $\overline{E_1}$ against the number of retained KL modes. For every correlation length the effective variability falls below the scalar baseline, and the shorter the correlation the stronger the attenuation.

for PCE degree 1, 3.0×10^{-13} MPa for degree 2, 7.3×10^{-4} MPa for the Gaussian Process and 3.6×10^{-1} MPa for the network; Fig. 10 shows the full convergence behaviour.

What this experiment can and cannot support deserves to be stated plainly. The reference response is itself a degree-2 Hermite polynomial, so a degree-2 expansion lies in a model class that contains it exactly; the $O(10^{-13})$ figure measures exact representability and the conditioning of the regression, not generalisation error, and it must not be read as PCE being twelve orders of magnitude more accurate than a network in any general sense. The factor of roughly 500 that is meaningful is the one between the two data-driven surrogates. What the comparison does measure is the value of a correct structural prior under a severe data budget: given that the response really is smooth and near-linear, the polynomial basis extracts it from tens of samples, whereas the Gaussian Process must infer smoothness from data and the network the functional form as well. Had the reference contained a feature outside the polynomial class — a damage bifurcation, a threshold, a discontinuous derivative — the ranking could reverse.

The Gaussian Process additionally supplies an epistemic error estimate that the PCE does not. At $N = 66$ its posterior standard deviation is 0.008 MPa against an aleatory response standard

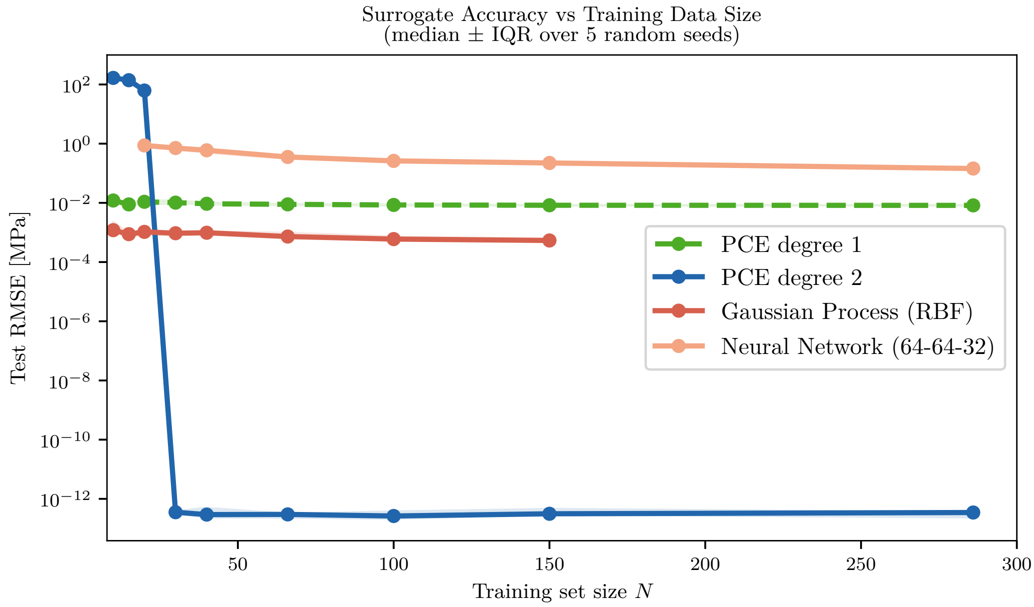


Figure 10: Test RMSE against training-set size for the four surrogates, median and interquartile range over five seeds. The network exhibits the classic overfitting signature, its error growing as the training set is enlarged.

deviation of 2.37 MPa, that is 0.3%: surrogate error is negligible beside the physical scatter it is meant to characterise. This is an estimate of epistemic uncertainty *within* the Gaussian Process model class and does not capture model-form error. For the PCE itself the corresponding evidence is the leave-one-out error, below 10^{-3} MPa against a nominal 210 MPa, together with the agreement between degree 2 and degree 3 reported in Table 2.

6. Discussion

6.1. Why E_1 dominates

For a response smooth in the inputs the output variance is approximated to first order by $\mathrm{Var}(Y) \approx \sum_i (\partial Y/\partial \xi_i)^2 \sigma_{\xi_i}^2$, so each parameter contributes through the product of its sensitivity and its variance, not through either alone. The panel carries the design pressure predominantly by in-plane membrane action, for which both the peak skin stress and the global deflection are set by the axial laminate stiffness $E_1 t$. The response is therefore close to linear in E_1, giving a large derivative, while E_1 simultaneously carries a non-negligible 5% scatter. The shear modulus is the instructive counter-example: its 10% scatter is twice as large, but the membrane-dominated kinematics make its derivative small, so the larger input variability is suppressed in the output. The ranking is governed not by which parameter is most uncertain but by which uncertainty the structure is configured to amplify.

6.2. Why SDEG is identically zero

That the interface remains undamaged across all 352 sampled configurations is a physical result with a methodological consequence. The design-point pressure lies comfortably within the elastic envelope of the adhesive; at nominal properties the interface traction stays below t_n, and even at the truncation bounds on K_n, t_n and G_{Ic} it never reaches the damage-initiation criterion. Damage is therefore not merely improbable but absent from the entire sampled domain. The three cohesive parameters are consequently *structurally unidentifiable*: because they never alter any response, no amount of additional sampling could assign them a non-zero Sobol index, and the nominally five-dimensional problem collapses, at this load, to an effectively one-dimensional problem in E_1. This is a finding rather than a defect — it tells the designer that the bond is not load-limiting here — but it also means that a study aimed at the cohesive law would have to drive the interface into damage.

6.3. Limitations

Several limitations delimit the regime in which the conclusions hold. The reported zero failure probability is bounded below by the resolution of the sampling and by the truncated input domain, as discussed in Section 4. The five inputs are treated as independent, whereas the elastic constants of a ply share common physical drivers and are in reality positively correlated; the proportional scaling of the dependent moduli simultaneously imposes perfect correlation on those quantities, an internally inconsistent simplification. Because E_1 dominates so completely the headline conclusions are robust to this, but a correlated input model, through a Nataf or Rosenblatt transform, would be required before the small G_{12} contribution could be read quantitatively. The study models a single representative panel rather than the complete fairing, so the absolute stress level should be regarded as representative of the coupon rather than as a certified margin. Only aleatory uncertainty in the material parameters is propagated; the epistemic error of the finite-element model itself — mesh discretisation, the mixed-mode law, the homogenised honeycomb — is not quantified. Finally, the KL study of Section 5.2 is confined to the input side, since its realisations were not propagated through the FE model.

None of these overturns the central finding, that fibre-direction modulus controls the structural variability and that the panel is reliable by a wide margin under the design load, but each points to a concrete extension: a correlated input model, a load level that activates the cohesive zone, mapped random-field realisations, and a full-fairing model with quantified discretisation error.

7. Conclusion

Non-intrusive polynomial chaos with Smolyak sparse quadrature was applied to quantify how manufacturing variability propagates through an Abaqus model of an H3 fairing sandwich panel. A second-order expansion built from 66 simulations captures the response statistics completely: a third-order expansion costing 286 simulations reproduces the mean and standard deviation to four significant figures and is marginally less accurate out of sample. Degree 2 is therefore the correct model order for this smooth, near-linear response rather than an economy measure, and the distinction matters because it rests on the demonstrated curvature of the response surface and not on a cost trade-off.

The sensitivity analysis yields one actionable message. The fibre-direction modulus governs more than 99% of the variance in both peak stress and deflection, so tightening its quality control is by far the most effective lever for reducing structural variability, while tolerances on shear stiffness and on the adhesive parameters are, at this load, of little consequence. Under the design pressure the panel is reliable by a wide margin, with a peak stress at 17.5% of the CFRP strength and a cohesive interface that never leaves its elastic regime.

The three extensions sharpen rather than merely supplement this picture. Bayesian identification shows that the parameter dominating the variance is also the one that a handful of stress measurements can pin down, which is what tolerance allocation requires, though it reduces epistemic and not aleatory uncertainty and therefore leaves the reliability assessment unchanged. The non-Gaussian translation field secures positivity of the modulus and quantifies the correlation distortion that the transform introduces, which proves negligible at the 5% scatter of E_1 but would not be at the larger scatter of the interface parameters. The surrogate comparison confirms that a correct structural prior is worth more than model flexibility when only tens of simulations are affordable, without supporting any universal ranking of the methods. The principal limitation throughout is that the design load never initiates damage, which leaves the cohesive parameters unidentifiable; the natural continuation is to repeat the analysis at a load that activates the cohesive zone, under a correlated input model and with random-field realisations propagated through the solver, so that the reliability of the bonded interface and not only of the skin can be assessed.

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